

Adaptive Headlight System

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Objective

- To develop an automated headlight control system for vehicles that adjusts the brightness depending upon the incoming vehicles' or objects' light or distance to reduce the risk of collision and temporary blindness.
- Implement the system using STM32 and Vega Boards and establish communication between the boards.

Problem Statement / Motivation

→ Night driving is risky by itself due to higher chances of accident at high speeds. It becomes even more dangerous if incoming vehicles' light is very bright or flashed suddenly, causing temporary blindness to the driver. In many cases, the lower headlight beam is not used despite the option.

→ Thus, there exists a space to automate brightness control of headlights such that the driver is not blinded by the lights and can drive comfortably at night.

Proposed Solution

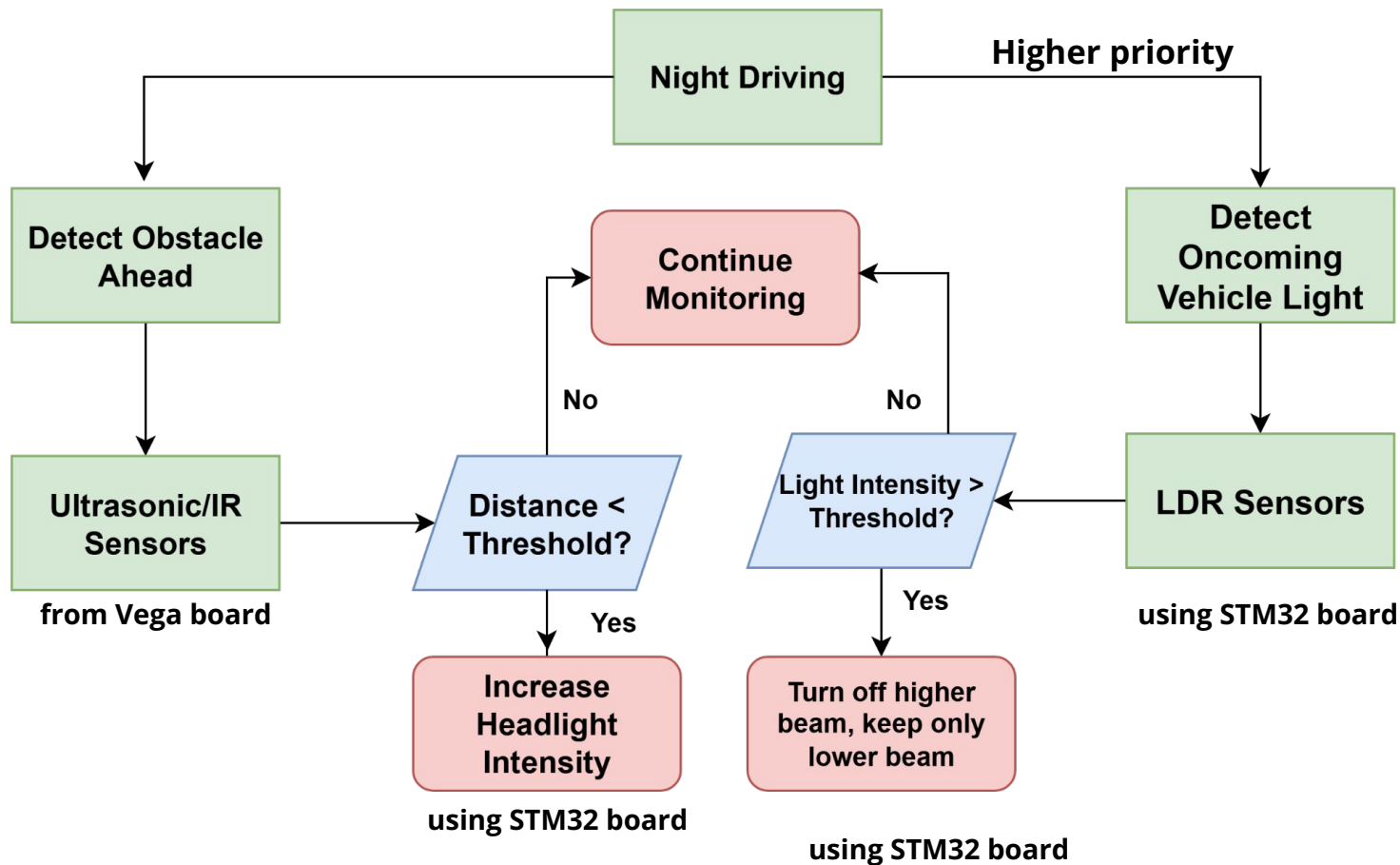
- There are two situations: there is an obstacle or an incoming vehicle (from opposite direction) ahead.
- There have to be sensors that detect the distance of the incoming object before reducing or increasing the brightness for safety.
- There is also a need of light sensors, specifically for incoming vehicles, for early detection of light and reduction of brightness of the driver car's headlights for the other person's safety.
- The sensors need to be in multiple directions

Approach towards the design

- For distance measurements, infrared (IR) sensors, or ultrasonic(HC-SR04) sensors could be used
- For light detection, LDR sensors can be a probable choice
- The control could be distributed between the two boards: STM32 for brightness control, and Vega board for sensing light and distance
- LED strips could be used for modeling the headlight. It needs to fade in or out while increasing or decreasing brightness instead of abrupt changes

More features to be added as an option

- Normalization: there could be an LDR sensor at the top of the vehicle for street lights, and the captured value could be used to normalize the output of LDR sensors at the front
- The headlight brightness could also be made sensitive to street lights: lower intensity on a well lit road, higher when dark



THANK you